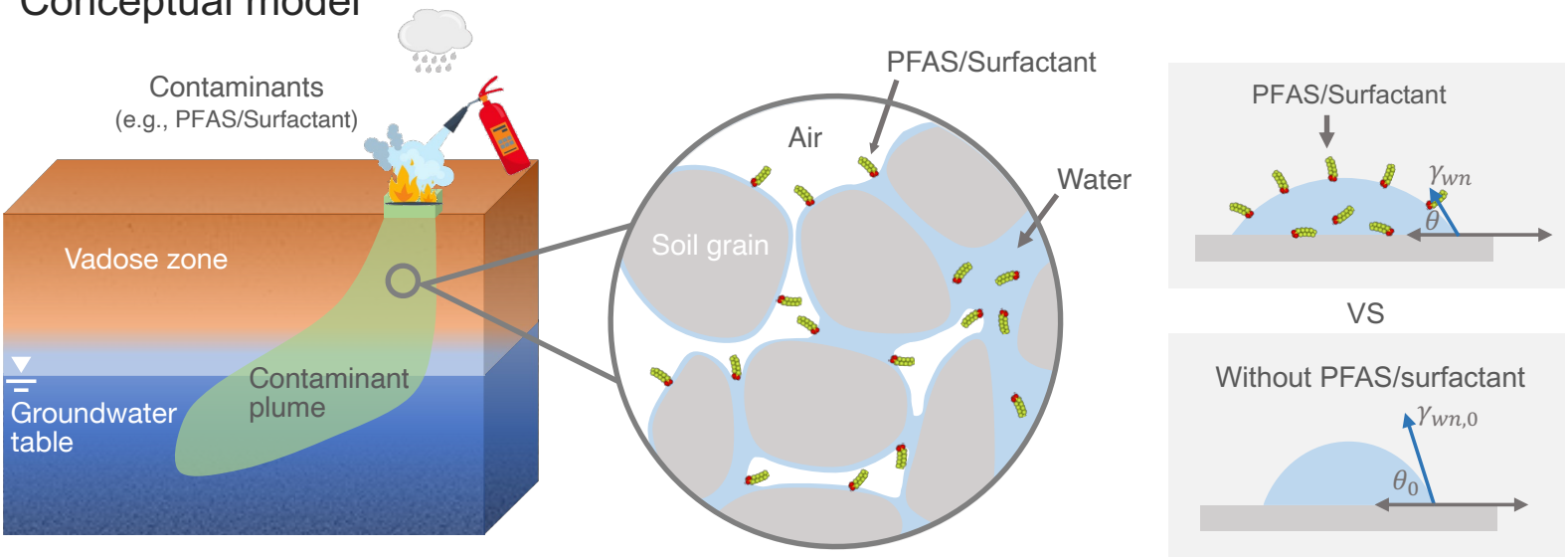
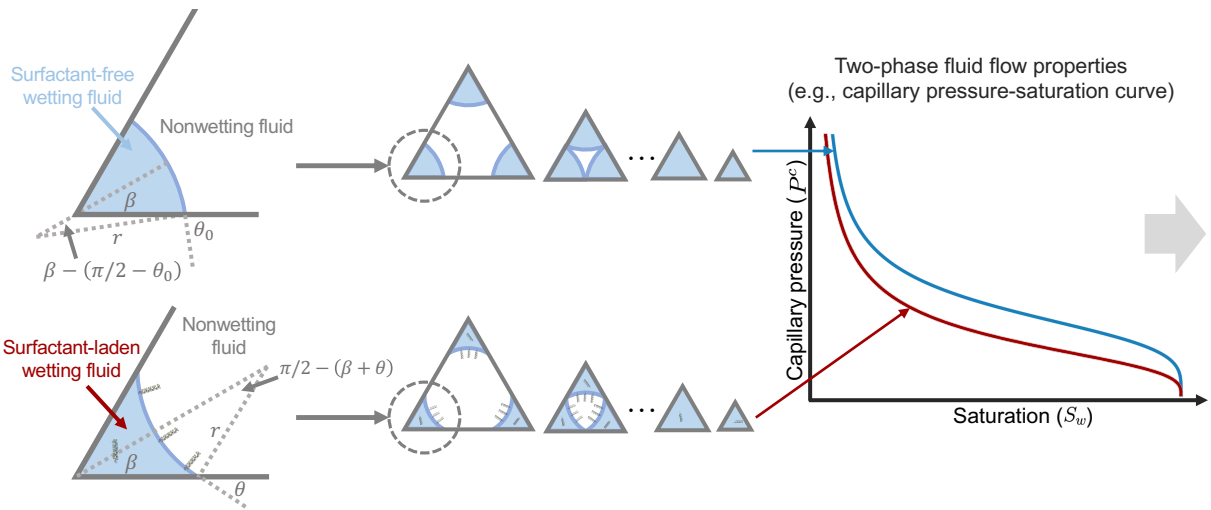
Conceptual model



Upscaling workflow



- Two-phase flow

$$\phi \frac{\partial S_\alpha}{\partial t} - \nabla \cdot \left[\frac{k_{r,\alpha}(S_w) k_0}{\mu_\alpha} \cdot \nabla \left(P_\alpha - \rho_\alpha g z \right) \right] = 0$$

$$P_{nw} - P_w = P^c(S_w)$$

- PFAS or Surfactant transport

$$\frac{\partial \phi A_{wn}(S_w) \Gamma_{wn}}{\partial t} + \sum_\alpha \frac{\partial (\phi S_\alpha C_\alpha + \phi A_{s\alpha}(S_w) \Gamma_{s\alpha})}{\partial t} + \sum_\alpha \nabla \cdot (q_\alpha C_\alpha - \phi D_\alpha \nabla C_\alpha) = 0$$

Note: $\alpha = w$ or nw, where "w" and "nw" refer to the wetting and nonwetting fluids, respectively.